

Data Structure and Algorithms

CSIT — 3rd Semester — 2079 — Answer Sheet
Subject Code: CSC211

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Why do we need to balance the binary search tree? Justify with an example. Create an AVL tree from the data 24, 12, 8, 15, 35, 30, 57, 40, 45, 78.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How recursive algorithm use stack to store intermediate results? Illustrate with an example. Convert the infix expression $A+B - (C*D/E+F) - G*H$ into postfix expression using stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How do you insert and delete a node at kth position of the doubly linked list? Describe the process of implementing stack and queue using linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Sort the numbers 82, 73, 12, 39, 26, 88, 2, 9, 60 and 41 using shell sort.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Why do we need asymptotic notation? Describe about Big Oh notation with its curve.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define a queue. Explain about enqueue and dequeue operation in circular queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Write a program to implement binary search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Find the MST of following graph using Prim's algorithm

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Assume you have to store the data {0,1,2,4,5,7} into a hash table of size 5, with hash function $H(x) = x \% 5$. Apply linear probing and double hashing as collision resolution techniques.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. In which case, the position of pivot element in quick sort is always either in first or last position. Create a max heap from the numbers {10,12,53,34,23,77,59,66,5,8}.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Evaluate the postfix expression $574-^*8/4+$ using stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a. Priority Queue b. Breadth First Traversal of a graph.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.